ID Y04-MA-W31-AK rev 345ad2

Number Detectives — multiples, factors and patterns with fractions and decimals

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Worksheet

Part A

a) 420 b) 14:45 c) 944 d) 36 (48 ÷ 4 = 12, 12 × 3 = 36) e) 8 f) obtuse
Spaced retrieval: Term 1 (Weeks 3, 9), Term 2 (Weeks 11, 13), Term 3 (Weeks 22, 23). Note any slips for the Week 39 revision.

C1

1. a) 7, 14, 21, 28, 35, 42, 49, 56 b) 12, 24, 36, 48, 60, 72, 84, 96 c) 54, 63, 72, 81, 90, 99
2. Circles: 4, 8, 12, ..., 60. Crosses: 6, 12, 18, ..., 60.
a) **12, 24, 36, 48, 60** b) **12** c) They go up by 12 each time — the common multiples are the multiples of 12 (the smallest common multiple).
3. a) 1 × 18, 2 × 9, 3 × 6 b) 1 × 20, 2 × 10, 4 × 5 c) 1 × 36, 2 × 18, 3 × 12, 4 × 9, 6 × 6 d) 1 × 17 only (17 is prime — preview of Part E!)
4. a) Array A: 2 × 12 (or 12 × 2); Array B: 3 × 8 (or 8 × 3) b) the others are 1 × 24 and 4 × 6 → **4 factor pairs** c) 1, 2, 3, 4, 6, 8, 12, 24
5.

Number	Digit sum	÷ 2	÷ 3	÷ 5	÷ 9	÷ 10
135	9		✔	✔	✔	
248	14	✔				
1 260	9	✔	✔	✔	✔	✔
3 474	18	✔	✔		✔	
5 005	10			✔		
7 299	27		✔		✔	

Watch: 7 299 looks “odd and unfriendly” but its digit sum 27 shows it divides by 9 (7 299 = 9 × 811).

C2

6. a) Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. Common factors of 24 and 30: **1, 2, 3, 6**; greatest **6**.
b) Factors of 16: 1, 2, 4, 8, 16; factors of 40: 1, 2, 4, 5, 8, 10, 20, 40 → common **1, 2, 4, 8**; greatest **8**.
7. a) A = ¾ B = 1½ (accept 1 ¾) C = 2¼ D = 2¾
b) 2⅓, 2⅔, 3 c) 4, 3½, 3
8.

Pattern	Next two	Rule
a)	3¾, 4¼	add ½
b)	4, 3⅔	subtract ⅓
c)	1 (or 1.0), 1.25	add 0.25
d)	3.0 (or 3), 3.1	add 0.1
e)	2 (or 2.0), 2.05	add 0.05
f)	5.55, 5.3	subtract 0.25

Watch d) and e): students who write 2.10 after 2.9, or 1.100 after 1.95, are treating the decimal part as a whole number.
9. a) 1.4, 2.0 (add 0.2) b) 2, 2½ (add ½) c) 0.4, 0.65 — the gap 0.9 − 0.15 = 0.75 is three equal jumps, so each jump is 0.25 d) 4.4, 5.0 — from 4.7 to 5.3 is 0.6 in two jumps, so each jump is 0.3.
10. a) add 0.05 b) stone 3 = 0.95, stone 5 = 1.05, stone 6 = 1.1, stone 8 = 1.2
c) Stone 20 is 19 jumps after stone 1: 19 × 0.05 = 0.95, and 0.85 + 0.95 = **1.8**. (Or: 20 × 0.05 = 1, then subtract the 0.2 "missing" at the start: stone n = 0.8 + 0.05 × n.)

C3

11. a) **True** — 6 = 2 × 3, so every multiple of 6 is 3 × (something). b) **False** — 9 (or 3, 15, 21, ...) is a multiple of 3 but not of 6; only *even* multiples of 3 are multiples of 6. c) **False** — the digit sum must divide by 9; 12 doesn't (e.g. 48 has digit sum 12, 48 ÷ 9 = 5 r 3). It *does* divide by 3, though.
12. 72. Multiples of 9 between 50 and 100: 54, 63, 72, 81, 90, 99. Even: 54, 72, 90. Multiples of 4: only 72 (54 ÷ 4 and 90 ÷ 4 leave remainder 2).
13. Digits without the box: 4 + 7 + 2 = 13. a) **5** (13 + 5 = 18 → 4 752) b) **2, 5, 8** (sums 15, 18, 21) c) **No** — the ones digit is 2, and a number divisible by 5 must end in 0 or 5. The box is the tens digit, so it cannot change that.
14. a) 16 → 5 factors (1, 2, 4, 8, 16); 25 → 3 (1, 5, 25); 36 → 9 (1, 2, 3, 4, 6, 9, 12, 18, 36) b) Each has a factor pair where a number is multiplied by **itself** (4 × 4, 5 × 5, 6 × 6). That pair gives only *one* new factor, so the total is odd. These are **square numbers** — they make square arrays.

Part D

15. Multiples of 6: 6, 12, 18, **24**; multiples of 8: 8, 16, **24** → a) **24** sausages and 24 rolls b) **4** packs of sausages (24 ÷ 6) and **3** packs of rolls (24 ÷ 8).
16. Common factors of 36 and 48: 1, 2, 3, 4, 6, 12 → a) **12** bags b) **3** apples (36 ÷ 12) and **4** muesli bars (48 ÷ 12).
17. Factor pairs of 48: 1 × 48, 2 × 24, 3 × 16, 4 × 12, 6 × 8. Rows of 4 to 12 chairs: **12 rows of 4, 8 rows of 6, 6 rows of 8, 4 rows of 12** (4 ways).
18. a) 3½ m, 5¼ m, 7 m b) 5 jumps = 8¾ m (not enough); **6 jumps** = 10½ m. *Shortcut:* 4 jumps = 7 m, so 8 jumps = 14 m — this helps students check.

Connections

X1. a)

Length	Width	Area	Perimeter
12	2	24	28
24	1	24	50
8	3	24	22
6	4	24	20

(Any order.) b) The length and width are a **factor pair** of 24 — length × width = area, so each rectangle is an array of 24 squares. c) **6 cm by 4 cm** (20 cm). It is the closest to a **square** — the more "square-ish", the shorter the perimeter; the long thin 24 × 1 has the longest perimeter.
X2. a) Multiples of 12: 12, 24, 36, 48, **60**; multiples of 15: 15, 30, 45, **60** → together every 60 minutes → **8:50 am**. b) 7:50, 8:50, 9:50, 10:50, 11:50 am → **5 times** (the next, 12:50 pm, is too late).
X3. a) Week 1 $7.45, Week 2 $9.90, Week 3 $12.35, Week 4 $14.80, Week 5 $17.25
b) Keep adding $2.45: Week 6 $19.70, Week 7 $22.15, Week 8 $24.60 (not enough), Week 9 **$27.05** → after **9 weeks**, with **$0.55** (55c) left over. *Shortcut:* $2.45 ≈ $2.50, so 4 weeks ≈ $10.
X4. a) ¾, 1½, 2¼, 3, 3¾, 4½ b) 4 batches = 3 cups, so 6 cups = **8 batches** c) 4 batches = 3 cups = 3 × 250 = **750 mL**.

Part E

19. a) **2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47** b) **15** c) The multiples of 11 below 50 are 22, 33 and 44; they are already crossed out as multiples of 2 or 3 (22 = 2 × 11, 33 = 3 × 11, 44 = 4 × 11). The next one, 55, is bigger than 50. *(Extension: any composite number up to 50 has a factor of 7 or less, because 8 × 8 = 64 > 50.)*
20. a) **composite** — 3 × 17 (digit sum 6) b) **prime** c) **composite** — 3 × 19 (digit sum 12) d) **composite** — 7 × 13 (the sneaky one!) e) **prime**
21. 3 & 5, 5 & 7, 11 & 13, 17 & 19, 29 & 31, 41 & 43 → **6 pairs**
22. 7 + 23, 11 + 19, 13 + 17 (**3 ways**). *Teacher note:* Goldbach's conjecture — every even number greater than 2 seems to be the sum of two primes; nobody has proved it yet.
23. No. 2 is prime (factors 1 and 2) and even. It is the **only** even prime, because every other even number has 2 as a factor, giving it at least three factors.

Fast-finisher bank

F1. a) 1, 2, 3, 4, 5, 6, 10, 12, 15, 20, 30, 60 → **12 factors** b) 18, 36, 54
F2. a) 8.85, 8.6 (subtract 0.25) b) 3, 2⅔ (subtract ⅓) c) **0.5** (10 × 0.05)
F3. by 2: **no** (ends in 9, odd); by 5: **no** (doesn't end in 0 or 5); by 9: **yes** — digit sum 1 + 2 + … + 9 = 45, and 45 divides by 9.
F4. 60 (smallest common multiple of 2, 3, 4, 5, 6 — note 4 × 3 × 5 = 60 already covers 2 and 6). Then **61** (one more than a common multiple).
F5. Common multiples of 6 and 8: 24, 48, **72**, …; 72 ÷ 9 = 8 ✔ → **72 seconds = 1 minute 12 seconds**.
F6. a) start at ¼, add ¾ b) 10th term = ¼ + 9 × ¾ = ¼ + 6¾ = **7** c) **No** — after 7 the next terms are 7¾ and 8½, so the pattern jumps over 8.
F7. 1 026 (digit sum 9; 1 008, 1 017 and 1 018 repeat digits). Strategy: start with 10_ _ and make the digit sum 9 with two new digits — 2 and 6 are the smallest that work.
F8. Prime: 2, 3, 5; composite: 4, 6; neither: 1. Prime is more likely: **⅗ (½)** compared with **⅔ (⅓)**.
F9. 48 with **10 factors** (1, 2, 3, 4, 6, 8, 12, 16, 24, 48). Runners-up: 36 (9 factors); 24, 30, 40, 42 (8 each).
F10. 4, 9, 25, 49. They are squares of prime numbers (2 × 2, 3 × 3, 5 × 5, 7 × 7): factors are 1, the prime and the number itself.
F11. a) **100** numbers (5 ÷ 0.05 = 100, or 20 numbers in every whole, × 5) b) **5** (1, 2, 3, 4, 5)
F12. a) Lockers **1, 4, 9, 16** b) Locker *n* is changed once by each student whose number is a factor of *n*. It ends open only if it is changed an odd number of times. Only square numbers have an odd number of factors (Q14), so only the square-numbered lockers stay open.

Homework

1. a) 8, 16, 24, 32, 40, 48, 56, 64, 72, 80 b) 24, 48, 72, 96
2. 1 × 40, 2 × 20, 4 × 10, 5 × 8
3. 1, 3, 9; greatest **9**
4.

Number	Digit sum	÷ 2	÷ 3	÷ 5	÷ 9	÷ 10
270	9	✔	✔	✔	✔	✔
1 035	9		✔	✔	✔	
4 128	15	✔	✔			

5. a) 2⅔, 3 — add ⅓ b) 4 (4.0), 3.75 — subtract 0.25 c) 0.45, 0.5 — add 0.05
6. a) 36 × 1 (P 74), 18 × 2 (P 40), 12 × 3 (P 30), 9 × 4 (P 26), 6 × 6 (P 24) b) **6 m × 6 m** (24 m of edging) — the square.
7. a) **Composite:** 8 + 7 = 15 divides by 3, and 87 = 3 × 29. b) **83** and **89**
8. 2 082 **9.** 211 r 1
10. 8:35 pm (6:40 + 2 h = 8:40, minus 5 min)
11. 0.07, 0.17, 0.7, 0.71
12. West
13. Prime (2, 3, 5 → 3 out of 6) is more likely than composite (4, 6 → 2 out of 6). 1 is neither.
14. a) 1, 13 and 25 December b) 12 days apart — 12 is the smallest common multiple of 3 and 4.
15. Model: "Factors of 12 are the numbers that divide into 12 exactly: 1, 2, 3, 4, 6, 12. Multiples of 12 are

the answers in the 12 times table: 12, 24, 36… Factors are smaller (or equal), multiples are bigger (or equal), and there are only six factors but the multiples go on forever."

Teacher notes

- **Common misconceptions:**
  - Mixing up *factor* and *multiple* (rhyme: "**F**actors **F**it into, **M**ultiples **M**ake more").
  - Missing factor pairs — insist on working up systematically from 1 until the pairs repeat; forgetting 1 × *n* and *n* itself.
  - Thinking 1 is prime (it has only one factor), or that 2 is not prime because it's even.
  - Applying the "digit sum" trick to 2, 5 or 10 (those depend on the last digit only), or to 4 and 6 (not taught yet).
  - Decimal patterns: writing 2.10 after 2.9 or 1.100 after 1.95 — the "whole-number thinking" error from Week 14. Have them say "two and nine tenths, three wholes" aloud or use a number line.
  - Fraction patterns: writing 4/4 or 1 ¾ instead of 1; not converting improper fractions to mixed numbers.
  - "Odd numbers are prime" — 9, 15, 21, 27, 51, 57 and 91 are good counter-examples.
- **If the student finishes early:** F12 (lockers) and F9 make good open investigations — extend F12 to 100 lockers, or ask "which number from 1 to 100 has the most factors?" (60, 72, 84, 90 and 96 tie with 12 each). For primes, extend the sieve to 100 (25 primes).
- **If the student struggles:** build arrays with counters or square tiles for factor pairs; use a hundred chart and coloured transparent counters for multiples; use a ruler/measuring jug marked in quarters for fraction patterns, and money (5c, 10c, 25c steps) for decimal patterns.
- **Sources / fact-check:** The Sieve of Eratosthenes is named after Eratosthenes of Cyrene (c. 276–194 BCE), so "about 2 200 years ago" is a safe round figure. Goldbach's conjecture (1742) is still unproven. An Australian metric cup is 250 mL. The bus timetable, savings amounts, prices and flashing lights are invented for the problems.
- **Next week builds on (Week 32 — unknowns and pre-algebra):** factor pairs → missing-factor number sentences (☐ × 6 = 48); pattern rules → input/output tables and inverse operations; Connections next week uses perimeter with a missing side written as ☐.